十一、 考虑 (r-1)次均未投到 6 且 r 次投到的概率

$\therefore p(r) = \left(\frac{5}{6}\right)^{r-1} \cdot \left(\frac{1}{6}\right) = \frac{5^{r-1}}{6^r}$

$\therefore E(X) = \sum_{r=1}^{\infty} \frac{5^{r-1}}{6^r} \cdot r = 6$

十三. $\because p(99) = \left(\frac{1}{6}\right)^3$

$p(10) = \left(\frac{1}{6}\right)^2 \cdot \frac{5}{6} \cdot 3$

$p(1) = \left(\frac{1}{6}\right) \cdot \left(\frac{5}{6}\right)^2 \cdot 3$

$\therefore E(X) = 0.875$

即要 0.875 元为公平的.

十五. 1° 设 (r-1)次均未取到 且r次取到；最多取(n-1)次

$p(r) = \frac{n-2}{n} \cdot \frac{n-3}{(n-1)} \cdots \cdot \frac{n-r}{n-r+2} \cdot \frac{2}{n-r+1}$

即 r次才打开的概率为 $\frac{2\cdot(n-r)\cdot(n-2)!}{n!} = \frac{2(n-r)}{n(n-1)}$

即 $E(X) = \sum_{r=1}^{(n-1)} r \cdot \frac{2(n-r)}{n(n-1)} = \frac{2}{n(n-1)} \cdot \left(n \cdot \frac{n(n-1)}{2} - \frac{n(n-1)(2n-1)}{6}\right)$

$= n - \frac{(2n-1)}{3} = \frac{n+1}{3}$

2° 若乱选,

则 $p(r) = \frac{(n-2)^{r-1}}{n^{r-1}} \cdot \frac{2}{n} = \frac{2(n-2)^{r-1}}{n^r}$

即 $E(X) = \sum_{r=1}^{\infty} r \cdot \frac{2(n-2)^{r-1}}{n^r} = \frac{2}{n} \cdot \frac{1}{\left(1-\frac{n-2}{n}\right)^2} = \frac{2}{n} \cdot \frac{n^2}{4} = \frac{n}{2}$

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十六. 即 $P(X) = C_n^x p^x (1-p)^{n-x}$

$\therefore E(Y) = P(0) - P(1) + P(2) \cdots\cdots$

$= \sum_{k=0}^{n} (-1)^k P(k)$

又 $\because E(X) = np$

$\therefore E(Y) = \sum_{k=0}^{n} (-1)^k \cdot k C_n^k \, p^k (1-p)^{n-k}$

又 $\because n(1+x)^{n-1} = \sum_{k=0}^{n} k \cdot C_n^k \cdot x^{k-1}$

即 $\sum_{k=0}^{n} (-1)^{k-1} \cdot (-1) \cdot k \cdot C_n^k \cdot \left(\frac{p}{1-p}\right)^{k-1} \cdot (1-p)^n \cdot \left(\frac{p}{1-p}\right) = -(1-p)^n \cdot n \cdot \left(\frac{p}{1-p}\right) \cdot \left(1 - \frac{p}{1-p}\right)^{n-1}$

$= -(1-p)^{n-1} \cdot np \cdot \frac{(1-2p)^{n-1}}{(1-p)^{n-1}} = -np(1-2p)^{n-1}$

十七. $\because \sum_{k=0}^{\infty} P(X=k) = C \cdot \sum_{k=0}^{\infty} \frac{a^k}{(1+a)^{k+1}} = \frac{C}{1+a} \cdot \left[1 + \left(\frac{a}{1+a}\right) + \cdots\right] = \frac{C}{1+a} \cdot \left(1 - \frac{a}{1+a}\right)^{-1}$

$= \frac{C}{1+a} \cdot (1+a) = C$； 即 $C = 1$

而 $EX = \sum_{k=0}^{\infty} P(k) \cdot k = \sum_{k=0}^{\infty} \frac{k \cdot a^k}{(1+a)^{k+1}}$

$\therefore \sum_{k=0}^{\infty} k \cdot x^{k-1} = \frac{1}{(1-x)^2}$； 即 $\sum_{k=0}^{\infty} k \cdot \left(\frac{a}{1+a}\right)^{k-1} \cdot \frac{a}{(1+a)^2} = \frac{1}{\left(1 - \frac{a}{1+a}\right)^2} \cdot \frac{a}{(1+a)^2} = a$

且 $E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot P(k) = \sum_{k=0}^{\infty} \frac{k^2 \cdot a^k}{(1+a)^{k+1}}$

$\therefore$

$\sum_{k=0}^{\infty} k^2 \cdot x^{k-1} = \frac{1+x}{(1-x)^3}$

$\therefore \sum_{k=0}^{\infty} \frac{k^2 a^k}{(1+a)^{k+1}} = \sum_{k=0}^{\infty} k^2 \cdot \left(\frac{a}{1+a}\right)^{k-1} \cdot \frac{a}{(1+a)^2} = \frac{1 + \frac{a}{1+a}}{\left(1 - \frac{a}{1+a}\right)^3} \cdot \frac{a}{(1+a)^2} = a \cdot (1+2a)$

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$\therefore D(X) = E(X^2) - E(X)^2$

$= 2a^2 + a - a^2$

$= a^2 + a$

十九.

(a) $\because E(X) = \sum_{k=0}^{\infty} k \cdot P(X=k)$

又 $\because P(X=k) =$ ~~$P(X \geq k) - P(X \geq$~~

$P(X \geq k) - P(X \geq k+1)$

$\therefore E(X) = \sum_{k=0}^{\infty} k \cdot [P(X \geq k) - P(X \geq k+1)]$

$= \sum_{k=0}^{\infty} P(X \geq k+1)$

$= \sum_{k=1}^{\infty} P(X \geq k)$

(b) $\because E(X^2) = \sum_{k=0}^{\infty} k^2 \cdot P(X=k)$

$= \sum_{k=0}^{\infty} (k^2 P(X \geq k+1) - k^2 P(X \geq k)]$

$= \sum_{k=0}^{\infty} (-k^2 + (k+1)^2) P(X \geq k+1)$

$= \sum_{k=0}^{\infty} (2k+1) P(X \geq k+1)$

$= \sum_{n=1}^{\infty} (2n-1) P(X \geq n)$

$\therefore D(X) = E(X^2) - E(X)^2$

$= 2\sum_{n=1}^{\infty} n P(X \geq n) - E(X)^2 - E(X)$

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二十.

∵[illegible].

$\because E(X) = \sum_{n=1}^{\infty} P(X \geq n)$

且 $P(X \geq 1) = 1$ ； $P(X \geq 2) = 1$

$P(X \geq 3) = p \cdot q \cdot 2$

$P(X \geq 4) = p \cdot q \cdot p + q \cdot p \cdot q = pq^2 + qp^2$

$P(X \geq 5) = 2 \cdot p^2 q^2$

$\vdots$

$\therefore E(X) = 1 + 1 + 2 \cdot (pq + p^2q^2 + \cdots) +$ [illegible] $(p+q)(pq + p^2q^2 + \cdots)$

$= 2 + 2 \cdot \frac{1}{1-pq} + (p+q)\frac{1}{1-pq}$

$= \frac{1}{1-pq} \cdot (2 + p + q) + 2$

二十一.

(a) $\because X = 1 \sim 6$, 且 $p = \frac{1}{6}$, 即 $E(X) = \frac{7}{2}$

$\therefore E(X) \cdot E(Y) = \frac{49}{4}$

且 $XY =$ 　　且 $p = \frac{1}{36}$

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2					
3	3					
4	4					
5	5					
6	6					

即 $E(XY) = \frac{21 \cdot 21}{36} = \frac{49}{4}$

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(b) $\because E(X) = \frac{7}{2}$； $E\left(\frac{1}{Y}\right) = \frac{49}{120}$

$\therefore E(X)\cdot E\left(\frac{1}{Y}\right) = \frac{343}{240}$

且 $\frac{X}{Y}$：考虑 $X=1$ 时，$\sum_{k=1}^{6} \frac{1}{k} = \frac{49}{20}$

$\therefore E\left(\frac{X}{Y}\right) = \frac{1}{36}\cdot\frac{49}{20}\cdot$ [illegible] $21 = \frac{343}{240}$

(c) $\because Y$：

Y	1	2	3	4	5	6
P	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$7-Y$：

$7-Y$	6	5	4	3	2	1
P	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

(d) $X+Y$：

$X+Y$	2	3	4	5	6	7	8	9	10	11	12
P	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{3}{36}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{3}{36}$	$\frac{1}{18}$	$\frac{1}{36}$

[illegible]

$7+X-Y$：

$7+X-Y$	2	3	4	5	6	7	8	9	10	11	12
P	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{3}{36}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{3}{36}$	$\frac{1}{18}$	$\frac{1}{36}$

故相同.

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